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Culture-Negative Infective Endocarditis Presenting as ANCA-Positive Glomerulonephritis

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Study Design A
Data Collection B
Statistical Analysis C
Data Interpretation D
Manuscript Preparation E
Literature Search F
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Patient: Male, 25-year-old
Final Diagnosis: Infective endocarditis due to *Bartonella henselae*
Symptoms: Abdominal pain • nausea and vomiting • night sweats
Clinical Procedure: —
Specialty: Cardiology • Infectious Diseases • Nephrology

Objective: Rare disease

Background: Infective endocarditis is challenging to diagnose when blood cultures are negative, and its symptoms are nonspecific. *Bartonella* species are fastidious intracellular bacteria that can cause culture-negative endocarditis and immune-mediated complications such as glomerulonephritis, often mimicking ANCA-associated glomerulonephritis.

Case Report: A 25-year-old man who underwent a Ross procedure for congenital aortic stenosis performed in childhood presented with 1 month of night sweats, abdominal pain, and intermittent nausea. Initial evaluation revealed elevated creatinine, pancytopenia, proteinuria, microscopic hematuria, and splenomegaly. Serologies showed positive c-ANCA and elevated PR3 antibodies. A presumed diagnosis of ANCA-associated glomerulonephritis led to initiation of high-dose steroids and planned rituximab. Kidney biopsy demonstrated crescentic glomerulonephritis with C3-dominant immune complex deposition, atypical for pauci-immune disease, prompting discontinuation of steroids.

Further infectious workup revealed high *Bartonella henselae* IgM and IgG titers. The patient met the criteria for possible infective endocarditis under the 2023 Duke-ISCVID criteria. Transesophageal echocardiography showed valvular abnormalities without definite vegetations, although only a prior transthoracic echocardiogram was available for comparison. The patient was treated with doxycycline and rifampin for 12 weeks, resulting in clinical and laboratory improvement.

Conclusions: This case demonstrates that *Bartonella* endocarditis can closely mimic ANCA-associated glomerulonephritis and rapidly progressive glomerulonephritis. In patients with systemic symptoms, positive ANCA serologies, and negative cultures, occult infection must be a key consideration, especially in patients at high risk for infective endocarditis. A careful exposure history, including pets, travel, and dental procedures, can provide crucial diagnostic clues. Early recognition and targeted antimicrobial therapy are essential to prevent irreversible organ damage and avoid inappropriate immunosuppression.

Keywords: Antineutrophil Cytoplasmic Antibody-Associated Vasculitis • *Bartonella henselae* • Cardiovascular Infections • Echocardiography, Transesophageal • Endocarditis, Bacterial • Glomerulonephritis

Full-text PDF: https://www.amjcaserep.com/abstract/index/idArt/952252

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Introduction

Infective endocarditis (IE) is a microbial infection of the endocardial surface of the heart; most often the valves, resulting from colonization and proliferation of bacteria or fungi [1,2]. It is a complex disease and an important public health concern [3,4]. Gram-positive organisms, particularly streptococci, staphylococci, and enterococci, account for approximately 90% of cases, whereas gram-negative bacteria, including HACEK organisms (*Haemophilus*, *Aggregatibacter*, *Cardiobacterium*, *Eikenella*, and *Kingella*), fungi, and other fastidious organisms, are less-common etiologies [2].

Pathogenesis involves endothelial injury, typically from turbulent blood flow across abnormal valves, intracardiac devices, or injection-related trauma, which promote deposition of platelet-fibrin thrombi on the endocardial surface. Transient or persistent bacteremia then seeds these sterile deposits, and microorganisms with surface adhesins, such as *Staphylococcus aureus* clumping factors, attach and proliferate within this protected niche, forming vegetations [2].

While blood cultures often identify the causative organism and are part of the diagnostic criteria, culture-negative endocarditis is not uncommon. Often this is the result of prior antimicrobial treatment. However, it also may be due to a subset of fastidious organisms that are difficult or impossible to culture using standard techniques [2,5]. Among these organisms, *Bartonella* species; particularly *Bartonella henselae* and *B. quintana*, are recognized as important causes of culture-negative infective endocarditis [5-7].

These intracellular gram-negative bacilli are transmitted primarily through cat scratches or bites (*B. henselae*) or ectoparasites such as body lice (*B. quintana*) and are often associated with an indolent course with nonspecific symptoms and delayed diagnosis. Populations at higher risk include individuals with relevant exposures – cat contact for *B. henselae* or homelessness and body lice for *B. quintana* – as well as those with pre-existing valvular abnormalities, prior cardiac surgery, or immunocompromised states [4,5,8-11].

Clinically, *Bartonella* endocarditis often presents insidiously, with nonspecific systemic symptoms such as fatigue, weight loss, and low-grade fever, or may accompany systemic Bartonellosis [9,12]. Laboratory findings may include anemia, thrombocytopenia, elevated inflammatory markers, and, in some cases, renal involvement such as proteinuria or hematuria due to immune complex-mediated glomerulonephritis [13]. Because blood cultures are usually negative, diagnosis relies on serologic testing, polymerase chain reaction (PCR) assays, and echocardiographic evaluation. Transesophageal echocardiography may reveal subtle vegetations or valvular regurgitation, although prior valvular surgery can complicate interpretation [14-16].

According to the 2023 modified Duke-ISCVID criteria, a *Bartonella* IgG titer of ≥1: 800 is now considered a major criterion for the diagnosis of infective endocarditis [17]. This addition helps identify culture-negative cases in which vegetations may be absent or difficult to detect. In patients with systemic symptoms, predisposing cardiac conditions, and positive *Bartonella* serologies, the combination of major and minor criteria allows for classification as possible or definite endocarditis, facilitating timely diagnosis and management [17].

Bartonella endocarditis can also mimic autoimmune or vasculitic processes presenting with positive antineutrophil cytoplasmic antibodies (ANCA), hypocomplementemia, and renal dysfunction. Misdiagnosis can lead to inappropriate immunosuppressive therapy and subsequent complications [18-25]. Awareness of risk factors, detailed exposure history – including pets, travel, and recent invasive procedures – and a high index of suspicion are critical to early recognition and initiation of targeted antimicrobial therapy.

The following case illustrates these diagnostic challenges in a young patient with prior valvular surgery who initially presented with findings suggestive of ANCA-associated glomerulonephritis but was ultimately found to have culture-negative infective endocarditis due to *Bartonella henselae*. This case highlights the importance of maintaining suspicion for infectious etiologies in patients with systemic symptoms and autoimmune serologies to avoid delayed diagnosis and inappropriate immunosuppressive therapy.

Case Report

A 25-year-old man with a history of congenital aortic valve stenosis status after a childhood Ross procedure, major depressive disorder, tobacco use (½ pack per day), and recreational use of LSD and marijuana but no history of injection drug use presented to the Emergency Department with 1 month of generalized weakness, night sweats, nausea, non-bilious, non-bloody vomiting, dark-colored urine, and abdominal pain. He denied fever, chills, or other systemic symptoms.

On examination, he was alert and oriented ×3. Vital signs: BP 116/70 mmHg, HR 96 bpm, RR 16/min, SpO_2 96% on room air, temp 37.1°C, and weight 81.3 kg. Cardiac examination revealed a systolic murmur at the right second intercostal space. Pulmonary examination results were normal. Abdominal examination demonstrated mild epigastric tenderness without hepatomegaly. Physical examination was negative for splinter hemorrhages, Janeway lesions, Osler's nodes, and Roth spots.

Initial laboratory evaluation showed elevated creatinine (1.9 mg/dL, eGFR 49 mL/min/1.73 m^2, albumin 2.6 g/dL), pancytopenia (WBC 4000/µL, hemoglobin 9.3 g/dL, platelets 102×10^3/µL), and normal cardiac troponin I. His highest creatinine level was 2.24 mg/dL. Coagulation studies showed prothrombin time 14 s and international normalized ratio (INR) 1.2. Urinalysis revealed microscopic hematuria (73 RBCs/hpf), proteinuria (1+), and trace leukocytes (7 WBCs/hpf). Toxicology was positive for THC only. Hepatitis B and hepatitis C serologies were negative. Hepatitis A total antibody was positive. A fourth-generation HIV screen was positive. The Infectious Diseases Department was consulted and HIV 1- and 2-specific antibodies and RNA quantitative viral load were negative, suggesting a false-positive result. Blood and urine cultures were negative. CRP was elevated at 0.77 mg/dL. A computed tomography (CT) scan of the abdomen demonstrated splenomegaly (14.2×10×17.2 cm, homogeneous) with no other abnormalities.

Given the persistent elevation of creatinine despite intravenous fluids and supportive care, and an unclear etiology, the Nephrology Department was consulted on hospital day 3. Further serological testing was performed, and the patient was discharged on hospital day 5 with plans for an outpatient kidney biopsy and follow-up in the nephrology clinic.

However, the patient returned to the Emergency Department 4 days after discharge with persistent night sweats and intermittent nausea and vomiting. He was re-admitted for further evaluation.

By the time of readmission, additional laboratory results from the prior admission were available, revealing markedly reduced complement levels (C3, C4, and total CH50 <12.6 U/mL). Serologic testing demonstrated positive c-ANCA and elevated PR3 antibodies, while ANA, MPO, anti-GBM, and anti-dsDNA antibodies were negative.

Upon re-consultation with the Nephrology Department, a presumptive diagnosis of ANCA-associated glomerulonephritis was made. A renal biopsy was performed, and pulse dose corticosteroids were initiated with 1 g methylprednisolone daily for 3 days. Rheumatology Department consultation was obtained and the patient was planned to start rituximab.

Preliminary biopsy results demonstrated immune complex deposition atypical for pauci-immune glomerulonephritis suggestive of an infection-associated process, prompting discontinuation of steroids. Final pathology results revealed crescentic glomerulonephritis with C3-dominant immune complex deposition on immunofluorescence, and sparse mesangial deposits on electron microscopy, without basement membrane disruption or foot process effacement.

The Infectious Diseases Department was consulted again. Additional history revealed a recent dental procedure without antibiotic prophylaxis, travel to Mexico City, and close contact with a kitten adopted 6 months prior, with reported scratches. Repeat blood cultures remained negative.

A transthoracic echocardiography (TTE) was obtained, showing significant valvular abnormalities. A transesophageal echocardiogram (TEE) showed mild regurgitation across the aortic, mitral, and pulmonic valves, with turbulent flow but without definite vegetations (**Figure 1-3**).

Ultimately, serologic testing returned positive for *Bartonella henselae* IgM (1: 128) and IgG (>1: 1024). Despite the absence of vegetations on echocardiography, a presumptive diagnosis

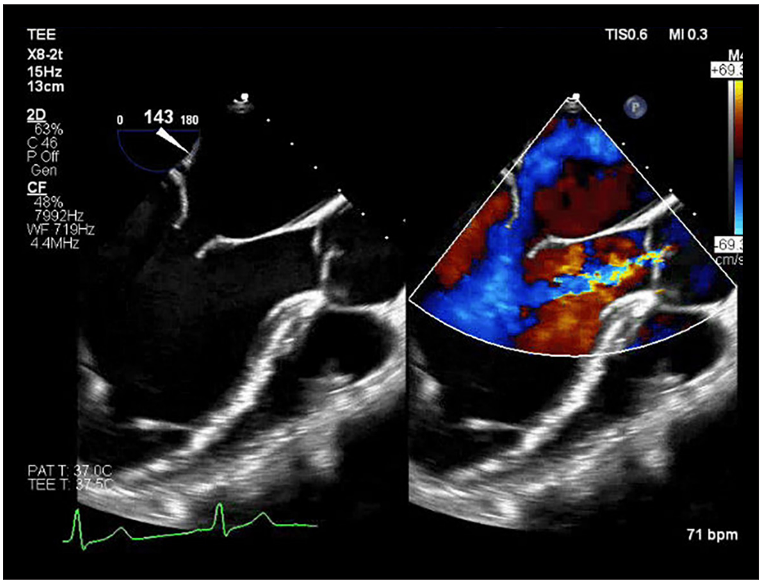

Figure 1. Transesophageal echocardiogram, mid-esophageal view showing mild aortic regurgitation. This image was taken from the patient's medical chart, and patient consent for publication was obtained.

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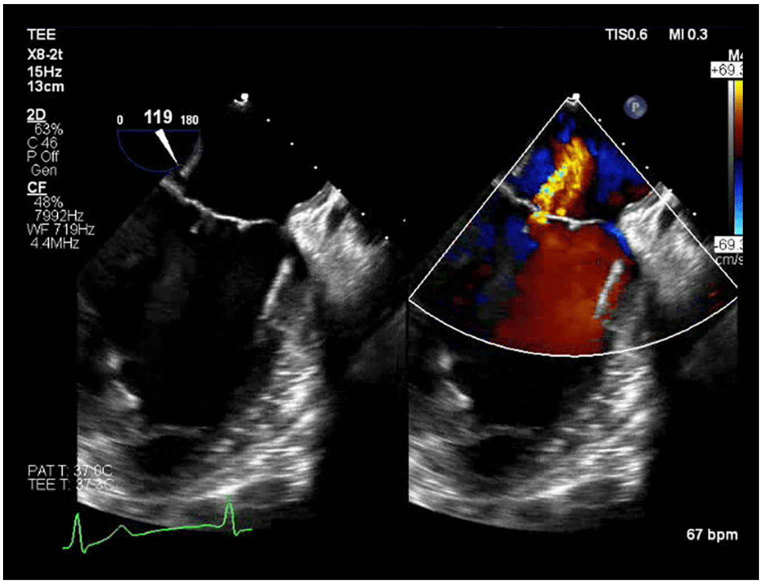

Figure 2. Transesophageal echocardiogram, mid-esophageal view showing mild mitral regurgitation. This image was taken from the patient's medical chart, and patient consent for publication was obtained.

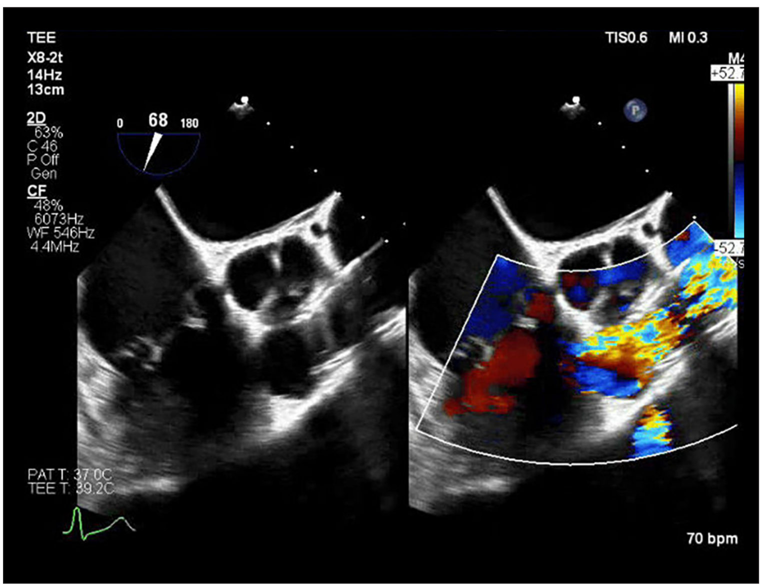

Figure 3. Transesophageal echocardiogram, mid-esophageal view showing pulmonary valve with turbulent blood flow. This image was taken from the patient's medical chart, and patient consent for publication was obtained.

of *Bartonella* endocarditis complicated by infection-associated ANCA-positive glomerulonephritis was made.

Ultimately, serologic testing returned positive for *Bartonella henselae* IgM (1: 128) and IgG (>1: 1024). According to the 2023 Duke-ISCVID criteria, a *Bartonella* IgG titer ≥1: 800 is a major microbiologic criterion for infective endocarditis. Minor criteria included the patient's history of congenital heart disease, previous valve repair, prosthetic valve, more than mild regurgitation, and complex-mediated glomerulonephritis. Given all the above and history of cat exposure with scratches, we had a strong suspicion for culture-negative infective endocarditis. Although transesophageal echocardiography did not demonstrate definitive vegetations, interpretation was further limited by the presence of multiple pre-existing valvular abnormalities and regurgitant lesions, making it difficult to determine whether any regurgitation represented a new change compared with prior studies. Because new valvular regurgitation can serve as an additional minor criterion under the Duke-ISCVID framework, the inability to clearly distinguish new from long-term valvular findings further complicated diagnostic classification. Therefore, the patient met the criteria for possible infective endocarditis and there was no better and unifying differential diagnosis that could explain the patient's constellation of findings. Therefore, a presumptive diagnosis of *Bartonella* endocarditis complicated by infection-associated ANCA-positive glomerulonephritis was made and the patient was started on oral doxycycline 100 mg twice daily and rifampin 300 mg twice daily for a planned 12-week course. He was discharged on day 7 of the second hospitalization. Doxycycline and rifampin were selected due to renal dysfunction and ease of oral administration, and a 12-week course was used to ensure adequate treatment of a patient with complicated history of heart disease.

Upon outpatient follow-up with the Infectious Diseases Department 6 weeks after hospital discharge, the patient reported marked improvement in his symptoms. He denied fever, rash, night sweats, or other new concerns. He stated he felt he had returned to near baseline. He continued to tolerate doxycycline and rifampin without adverse effects. Laboratory evaluation demonstrated normalization of his renal function (creatinine 1.2 mg/dL), resolution of cytopenias (WBC 6.2×10^3/µL, platelets 214×10^3/µL), and negative c-ANCA and p-ANCA. Repeat *Bartonella henselae* serologies showed decreasing IgG titers (1: 640) and negative IgM.

Discussion

Bartonella species are increasingly recognized as important causes of culture-negative infective endocarditis (IE), particularly in patients with predisposing cardiac abnormalities or relevant epidemiologic exposures. Infection with these organisms can trigger the formation of circulating immune complexes. These complexes can then deposit in the glomeruli and this process can cause consumption of complements, leading to hypocomplementemia. Also, neutrophil activation via infection and molecular mimicry can induce transient ANCA production. These immune-mediated processes can result in immune complex-mediated glomerulonephritis, mimicking ANCA-associated vasculitis, and highlight the potential for *Bartonella* endocarditis to have similar presentation to autoimmune disease [5,26]. This case was a diagnostic challenge because the presentation mimicked ANCA-associated glomerulonephritis, with renal involvement, cytopenias, and positive c-ANCA serology, ultimately leading to an initial trial of immunosuppression before the infectious etiology was identified. Such overlap has been described in the literature, where *Bartonella* endocarditis has been associated with false-positive ANCA serologies and immune complex-mediated glomerulonephritis, complicating the distinction between primary autoimmune disease and infection-driven secondary phenomena [20,22].

Some advanced echocardiographic modalities, such as pulsed-wave tissue Doppler imaging, have been explored as adjunct tools in the evaluation of intracardiac masses, including vegetations. The implementation of transthoracic echocardiography with pulse wave tissue Doppler imaging may improve visualization and differentiation of intracardiac masses via different color coding of the pathological structure compared to surrounding tissue. Also, this imaging modality can provide a detailed assessment of the specific pattern of motion of each intracardiac mass, with important implications. Although these advanced imaging techniques were not utilized in our patient, their application may provide additional diagnostic and prognostic information in cases where there is a suspected vegetation and conventional echocardiography yields inconclusive findings [27].

Several features make this case unique. First, the patient was a young adult with a prior Ross procedure, in whom transesophageal echocardiography revealed only mild regurgitation without vegetations, complicating interpretation. Echocardiographic findings in *Bartonella* IE are often subtle, and prior valve surgery or prosthetic material further decreases sensitivity, as reported in previous series [15,17]. Unlike other published cases where vegetations were eventually visualized, our patient never demonstrated definitive echocardiographic lesions, yet met diagnostic criteria based on high-titer *Bartonella* serology and clinical features. This underscores the importance of integrating serology and exposure history into the diagnostic approach, particularly when echocardiographic findings are equivocal.

Second, the renal manifestations were prominent and misleading. *Bartonella* endocarditis has been associated with glomerulonephritis due to circulating immune complexes, but the presence of c-ANCA positivity and PR3 elevation strongly suggested glomerulonephritis at presentation. Similar cases in the literature describe misdiagnosis leading to inappropriate immunosuppressive therapy, occasionally with adverse outcomes [28]. In contrast, our patient improved with discontinuation of steroids and initiation of targeted antimicrobial therapy, highlighting the reversible nature of *Bartonella*-associated renal disease when promptly recognized and appropriately treated.

Third, similar cases in the literature highlight the diagnostic challenges posed by *Bartonella* endocarditis, particularly in patients with prior valvular abnormalities or culture-negative presentations. Reported cases often describe patients with positive ANCA serologies, hypocomplementemia, or immune complex-mediated glomerulonephritis, which can mimic primary autoimmune disease and lead to inappropriate immunosuppression. In most reports, delayed recognition of *Bartonella* infection resulted in prolonged illness or the need for surgical intervention, whereas early serologic testing and targeted antibiotic therapy, as in our patient, allowed for clinical and renal recovery without surgery. These comparisons underscore the importance of maintaining a high index of suspicion for *Bartonella* in similar clinical scenarios and integrating epidemiologic history, serology, and imaging to guide management [10,12,15,25].

Finally, the patient's recovery following doxycycline and rifampin therapy reinforces current recommendations for combination therapy in *Bartonella* IE, especially in cases involving prior valvular surgery [12]. Our patient's favorable outcome

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contrasts with other reports in which delayed diagnosis led to progression of valvular destruction or need for repeat valve replacement. This case demonstrates that even in the absence of visible vegetations, early suspicion, appropriate serologic testing, and prompt antimicrobial therapy can avert surgical intervention and preserve organ function.

In summary, this case emphasizes the diagnostic complexity of *Bartonella* IE in patients with prior valve surgery and overlapping autoimmune features. Awareness of this entity, careful exposure history-taking, and judicious use of serologic testing are essential to avoid misdiagnosis, prevent unnecessary immunosuppression, and ensure timely initiation of effective therapy.

Conclusions

This case illustrates the diagnostic challenges of *Bartonella* endocarditis, particularly in patients with prior valvular surgery and systemic features mimicking autoimmune disease. Prompt recognition, integration of exposure history, and targeted serologic testing allowed for timely initiation of appropriate antimicrobial therapy, resulting in resolution of renal and hematologic abnormalities without the need for surgical intervention. The case highlights the importance of considering *Bartonella* in culture-negative endocarditis and underscores how early, accurate diagnosis can improve outcomes, avoid unnecessary immunosuppression, and contribute to the growing understanding of atypical presentations of this rare infection.

Institution Where Work Was Done

Northeast Georgia Medical Center, Gainesville, GA, USA.

Patient Consent

Obtained.

Declaration of Figures' Authenticity

All figures submitted have been created by the authors who confirm that the images are original with no duplication and have not been previously published in whole or in part.

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